

Project Brief Template

For students proposing their own project

PURPOSE

Model that detects defected toys in a toy factory. Toy type: teddy bear.

1. Project Identification

Decision / Question	Student Response
Student full name	Fazliddin Hamzayev
Project title	Defected product detector
Industry / field	Toy production
Intended client / user / stakeholder	The factory workers
One-sentence project summary	CNN model based on workflow of YOLO CV

2. Client / User Background

Describe the organization, user or context in which the problem exists. Explain only what is necessary to understand the project.

In some toy factories detection and inspection of defected products are still done by people. The model helps to replace human labor and decrease amount of load to the factory employees.

3. Business Problem

What is currently difficult, slow, expensive, inconsistent or impossible? Why does the problem matter?

More workers demand more money, which is not so profitable to managers. The model can help to minimize these emissions to zero.

4. Requested Solution

Describe the outcome you want the ML/AI system to provide. Do not describe the exact algorithm yet.

Detect defected product and send the exact product to recast.

5. Available / Expected Information

What types of data could realistically exist for this problem? Do not assume access to private data you cannot legally use.

YOLO – used in cameras to detect exact objects using computer vision (CV).

6. Data & Problem Discovery

Decision / Question	Student Response
Selected dataset and source	Google Images and the open Roboflow Universe teddy bear dataset — collected manually since there weren't many toy photos easily available in one place.
What does one record / sample represent?	512x512 photo of a teddy bear in one of 12 colors, labeled as “normal” or “defective” and type of defect, stored in labels.csv.
Proposed target or ML objective	Image classification that predicts whether toy is defected or not. Plus, defects are also classified as subclasses correspondingly.
Key information available at prediction / inference time	Photo of one toy (RGB Image).
Main data quality issues	As images come from different sources, background and lighting are not fully consistent across photos, which may affect model performance.
Potential leakage risks	As quality of some images is not so good, model may ignore minimal defects.
Privacy / fairness / licensing concerns	License: CC BY 4.0 (free to use, just attribute), https://universe.roboflow.com/teddybear/teddy-bears License: CC BY-NC-SA 4.0 , (non-commercial only), https://www.mvtec.com/company/research/datasets/mvtec-ad

7. Technical Proposal

Decision / Question	Student Response
ML problem formulation	Image classification with supervised learning method.
Proposed baseline	CNN, trained on the collected teddy bear image dataset.
Main modeling approach(es) to investigate	Object detection model as YOLO, to find defect.
Data splitting / validation strategy	Train/validation/split by label (type and color) to preserve class balance; validation set for model selection, held-out test set for final evaluation.
Primary evaluation metric(s) and why	Value fluctuates between 0 and 1; Toy includes defect – 0; Toy is normal – 1;
Expected inference input	A photo of a toy similar to the training images.

Expected inference output	Predicted label – normal or defected, with confidence score and defect type.
Main technical risks / assumptions	As training images come from different sources with varying quality, some predictions may be less reliable.

8. Functional Requirements

Write 4–7 outcome-focused requirements describing what the finished system must be able to do.

1. Accept a toy photo as an input.
2. Classify product as defected or normal.
3. Return accuracy, confidence score and predicted label.
4. Generalize across multiple colors of teddy bears in the training dataset.
5. Reach an agreed minimum recall on the “defective” on a held-out test set (threshold can be changed).

9. Expected Deliverables

- A working ML solution that addresses the approved problem.
- A trained model or ML pipeline with documented methodology.
- Documented data source and assumptions.
- Evaluation on appropriate unseen/held-out data.
- A usable inference interface or service appropriate to the project.
- A reproducible repository with clear run instructions.
- Documented limitations, risks and recommended next steps.

Note: General capstone requirements are defined in the official Capstone Evaluation Criteria and Helper documents.

10. Acceptance Criteria

Define what must be true for the project to be considered functionally complete.

- Hyperparameters can be changed to see how accurate model is
- Threshold can be changed to focus on exact things (especially useful while trying to find exact thing)
- The predictions or analysis process is reproducible from the submitted repository.
- The output is meaningful for the user use case described in this brief.

11. Constraints — In Scope / Out of Scope

IN SCOPE	OUT OF SCOPE
Binary classification (normal / defective) from a toy photo	Use in real production, as the model may still have accuracy drawbacks
Public data	A few defected toys can be considered as normal, so not that reliable
Model trained on the collected real photo dataset	Direct production integration

12. Questions You Must Resolve

List the key engineering/product questions that must be answered before or during implementation.

1. What minimum recall/precision on the defective class is acceptable to the client?
2. Which defect types are more likely to be detected first?
3. Where will the model run at inference time (cloud service, local script, mobile device)?

13. Optional Directions

List optional extensions only after the core scope is clear. These must not be required for the project to be considered complete.

- Multi-class defect-type classification_____
- Object-detection or possible segmentation to localize defect in the image_____
- Expanding the dataset with more real teddy bear photos from additional sources and conditions

14. Mentor Review & Approval

This section is completed by the mentor after reviewing the proposed scope and technical direction.

Decision	☐ Approved ☐ Approved with revisions ☐ Revision required
Required revisions / comments	
Approved scope / special conditions	
Mentor name	
Date	